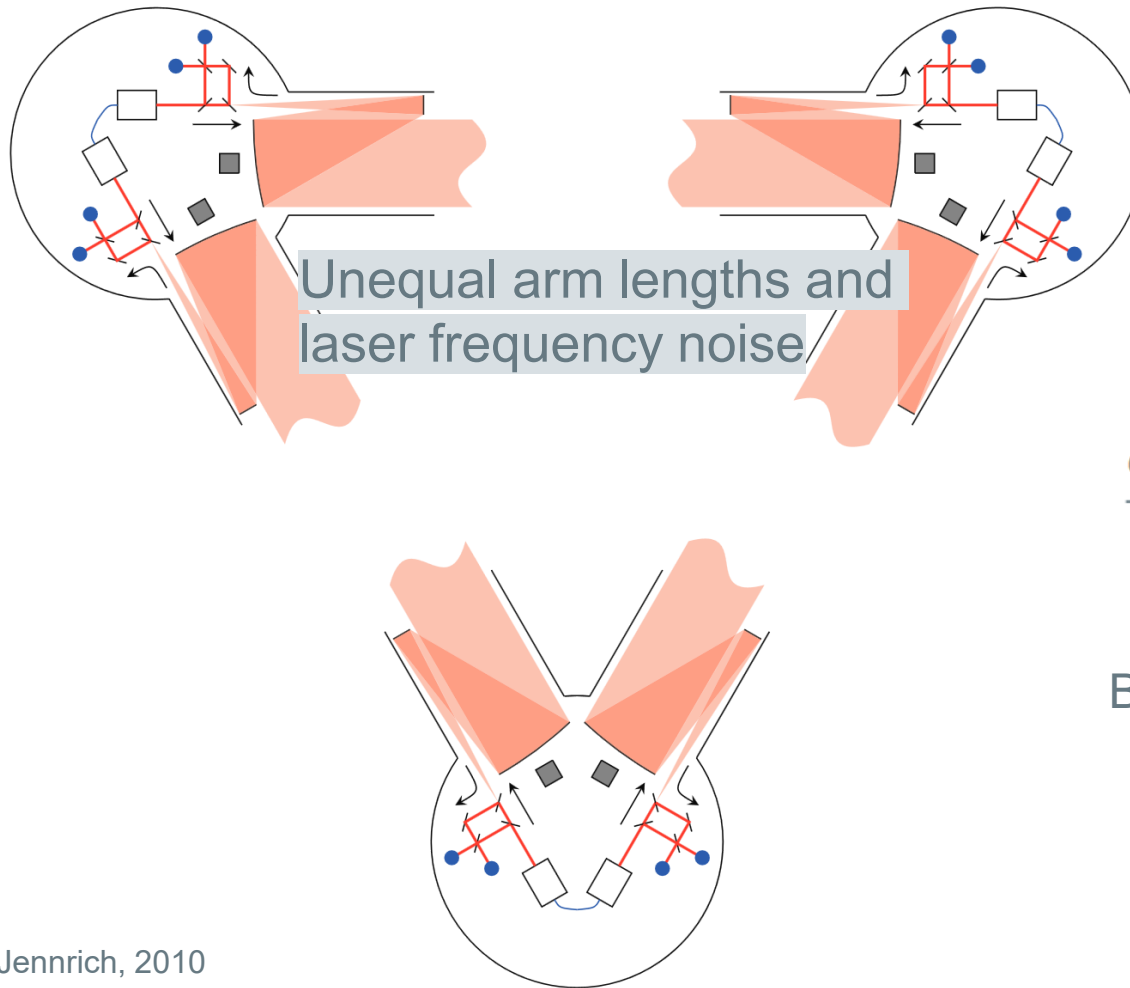




Testbed for Validating Second-Generation TDI and Clock Noise Correction for LISA

Karin Kruuse, Reid Ferguson, Patrick Krieger, Guido Mueller

The LISA Satellites



As per LISA requirements

$$\frac{\delta\nu}{\nu} = 10^{-13}$$

Hence for the arm length uncertainty we get

$$\frac{\delta\nu}{\nu} = \frac{\delta L}{\Delta L} \Rightarrow \delta L \approx 10^{-13} \cdot 10^7 \text{ m} = 1 \mu\text{m}$$

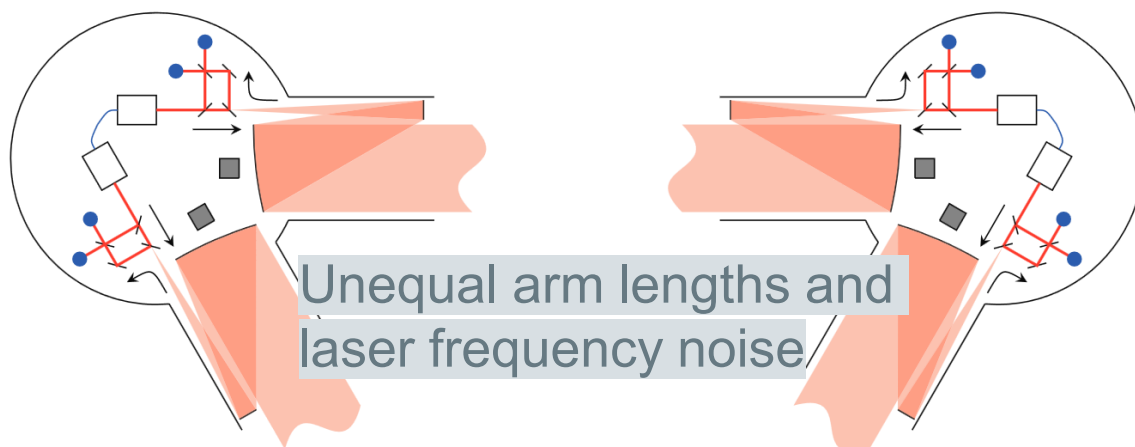
But to be sensitive enough to detect GWs, we need

$$\delta L = 10^{-13} \text{ m} \Rightarrow \Delta L = 1 \text{ m}$$

The LISA Satellites

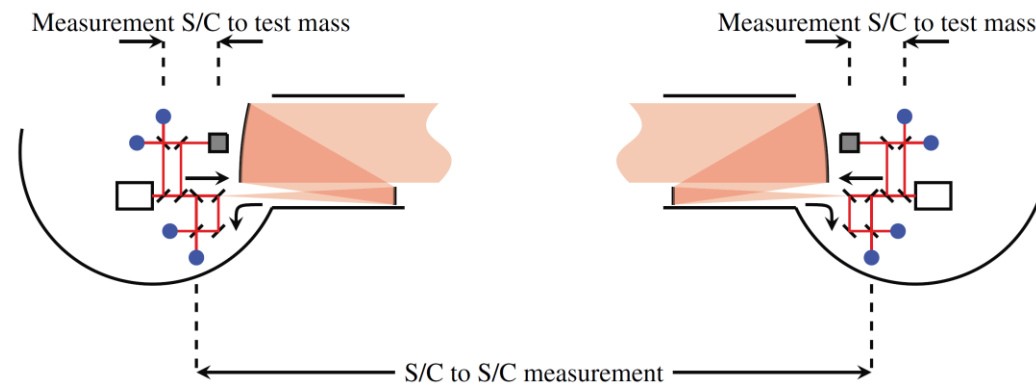
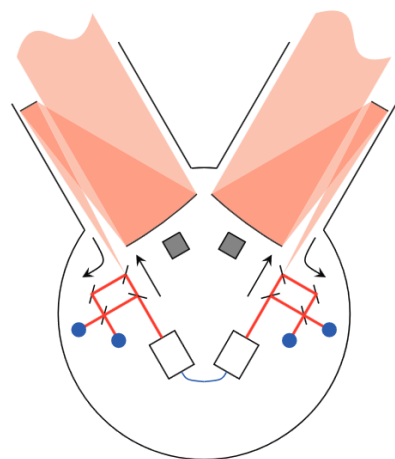


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Unequal arm lengths and
laser frequency noise

Split interferometry

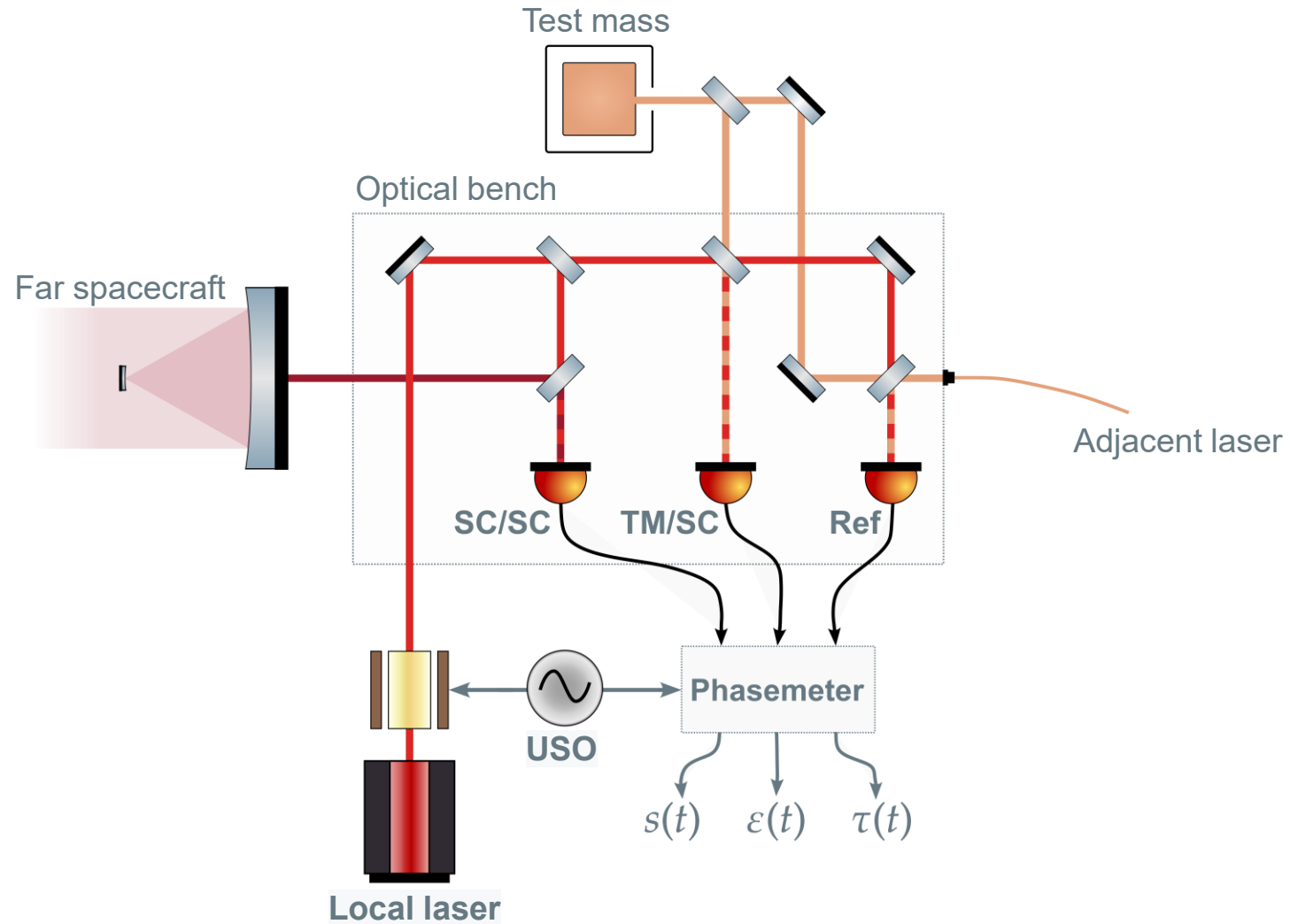


Jennrich, 2010

LISA Interferometric Metrology System



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USO – Ultra Stable Oscillator

$s(t)$ – science beat note

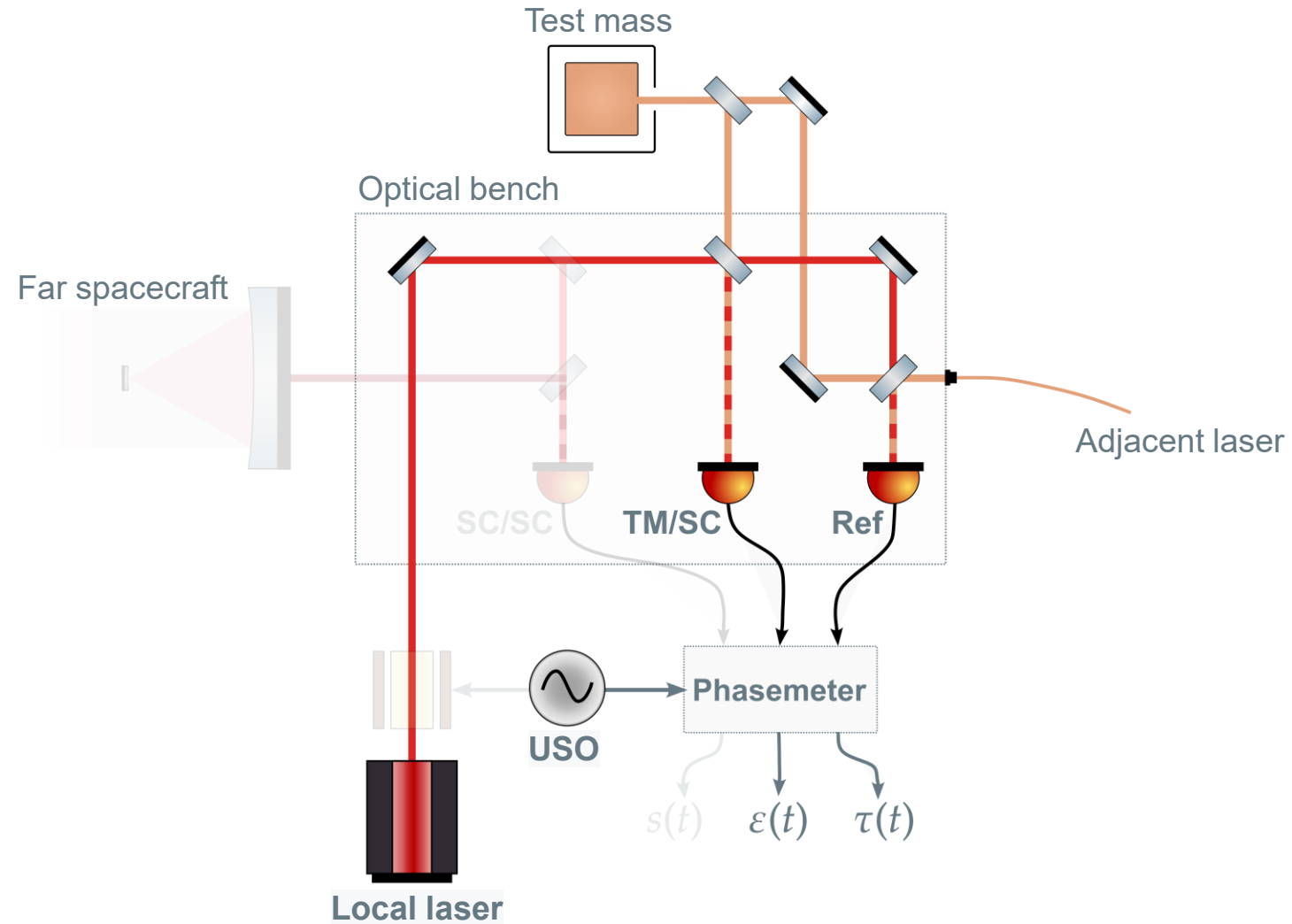
$\varepsilon(t)$ – test mass beat note

$\tau(t)$ – reference beat note

LISA Interferometric Metrology System



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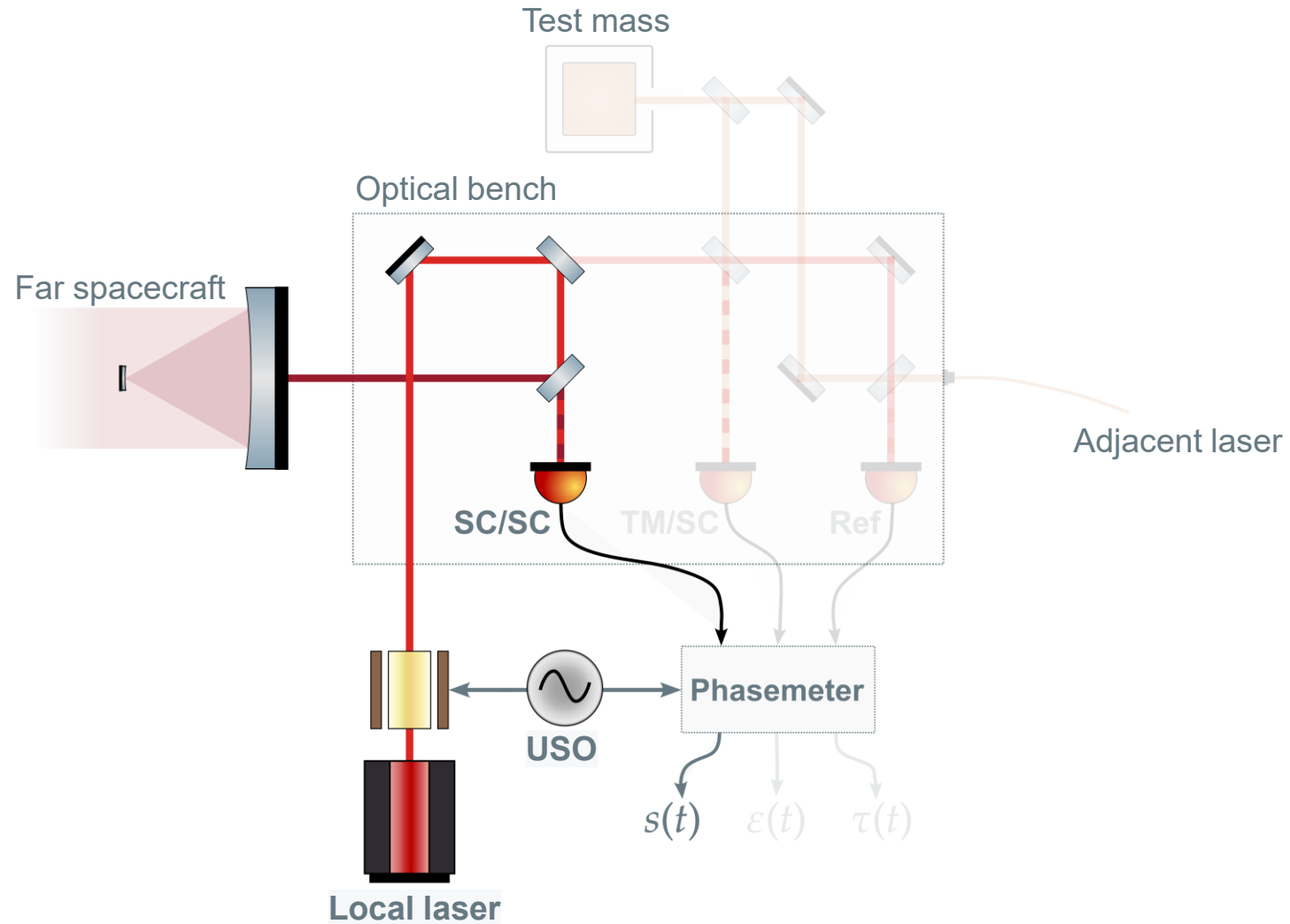
$\varepsilon(t)$ – test mass beat note

$\tau(t)$ – reference beat note

LISA Interferometric Metrology System



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USO – Ultra Stable Oscillator

$s(t)$ – science beat note

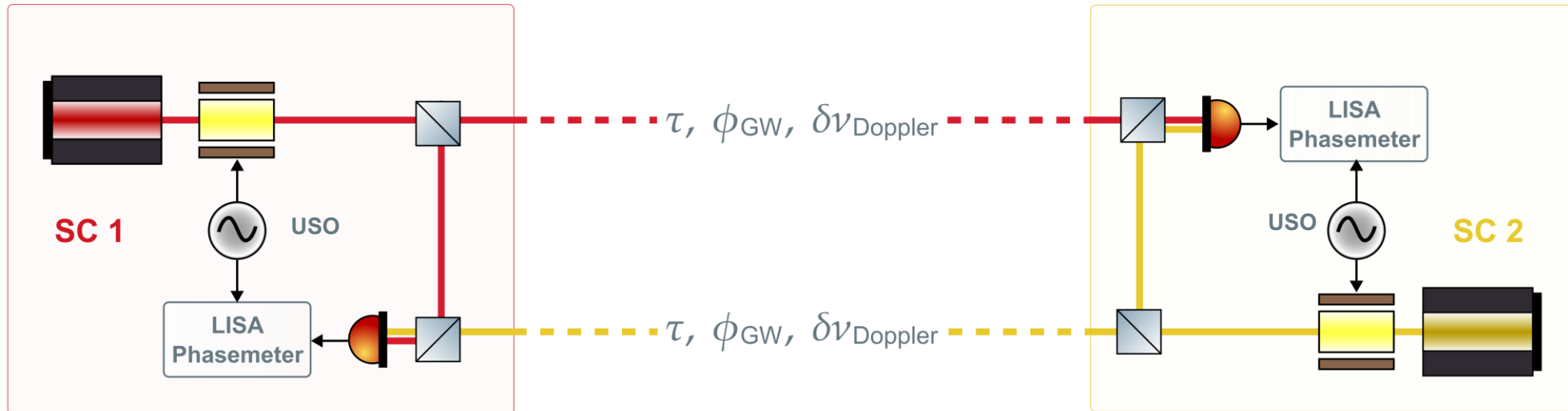
$\varepsilon(t)$ – test mass beat note

$\tau(t)$ – reference beat note

One-arm Phase Measurement



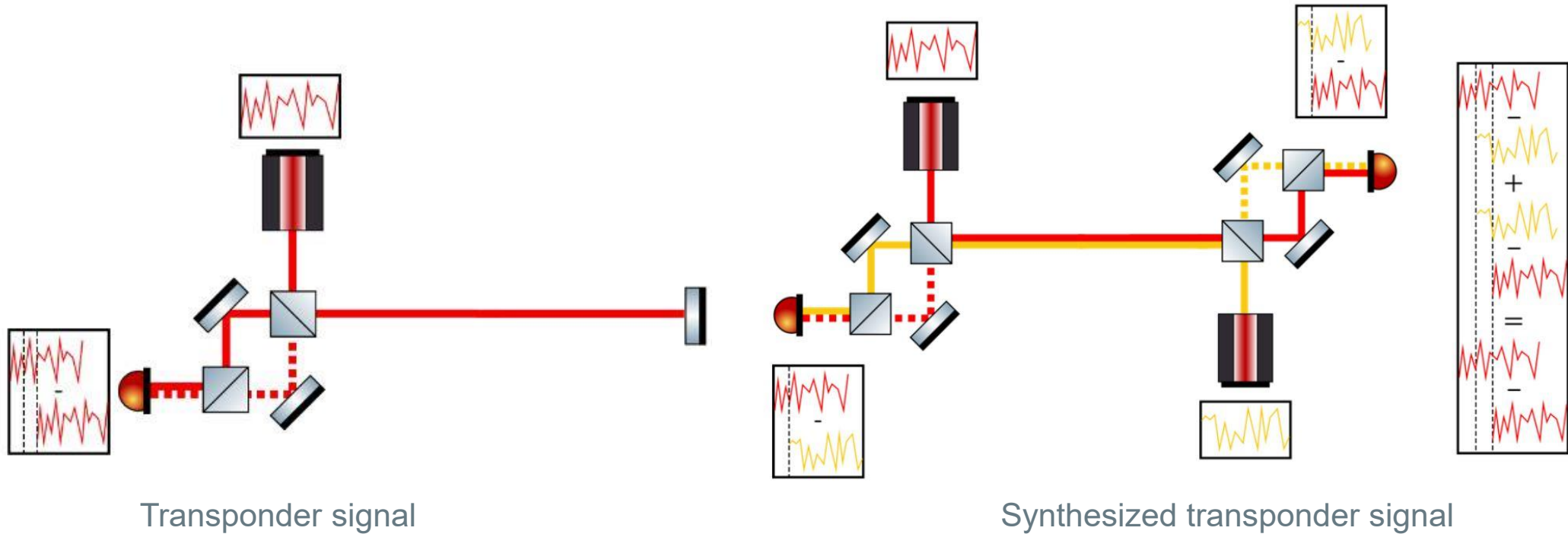
Bare-bones LISA arm



Laser Noise Reduction via TDI



Time Delay Interferometry (TDI) is a cluster of methods to construct virtual equal arms in post-processing.



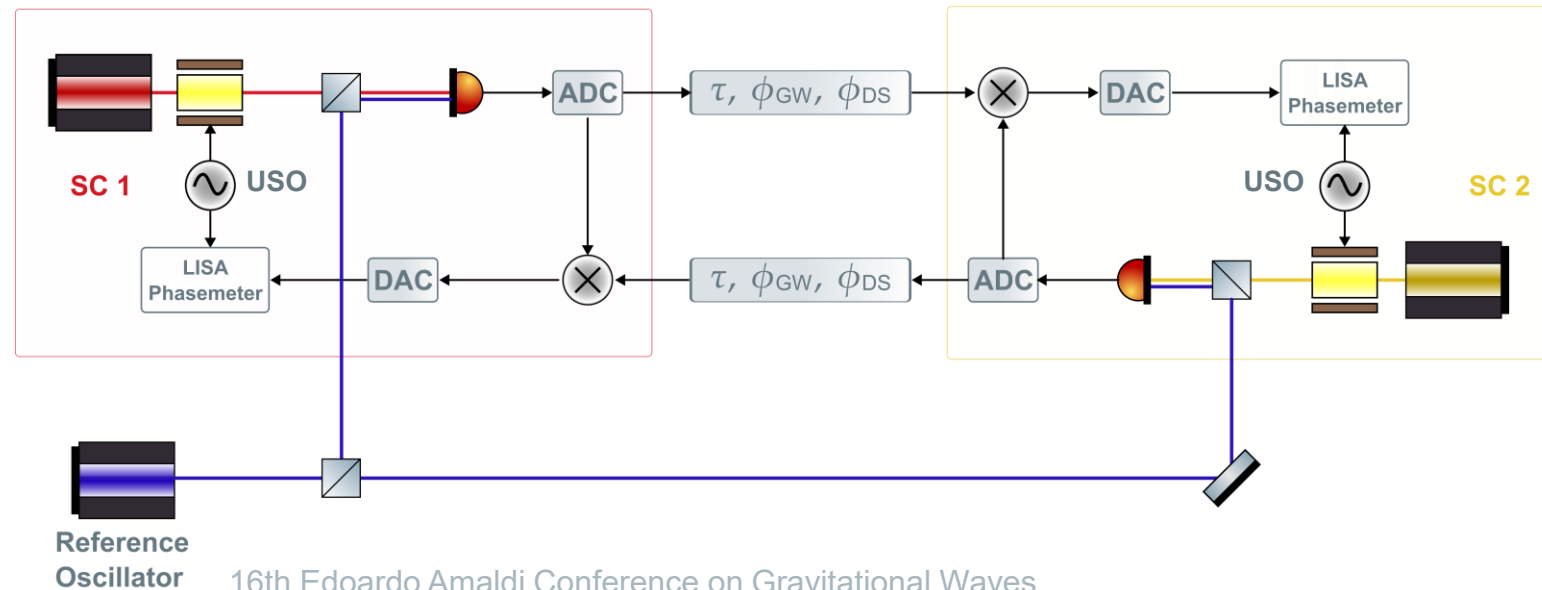
One-arm Phase Measurement



LISA arm



testbed arm



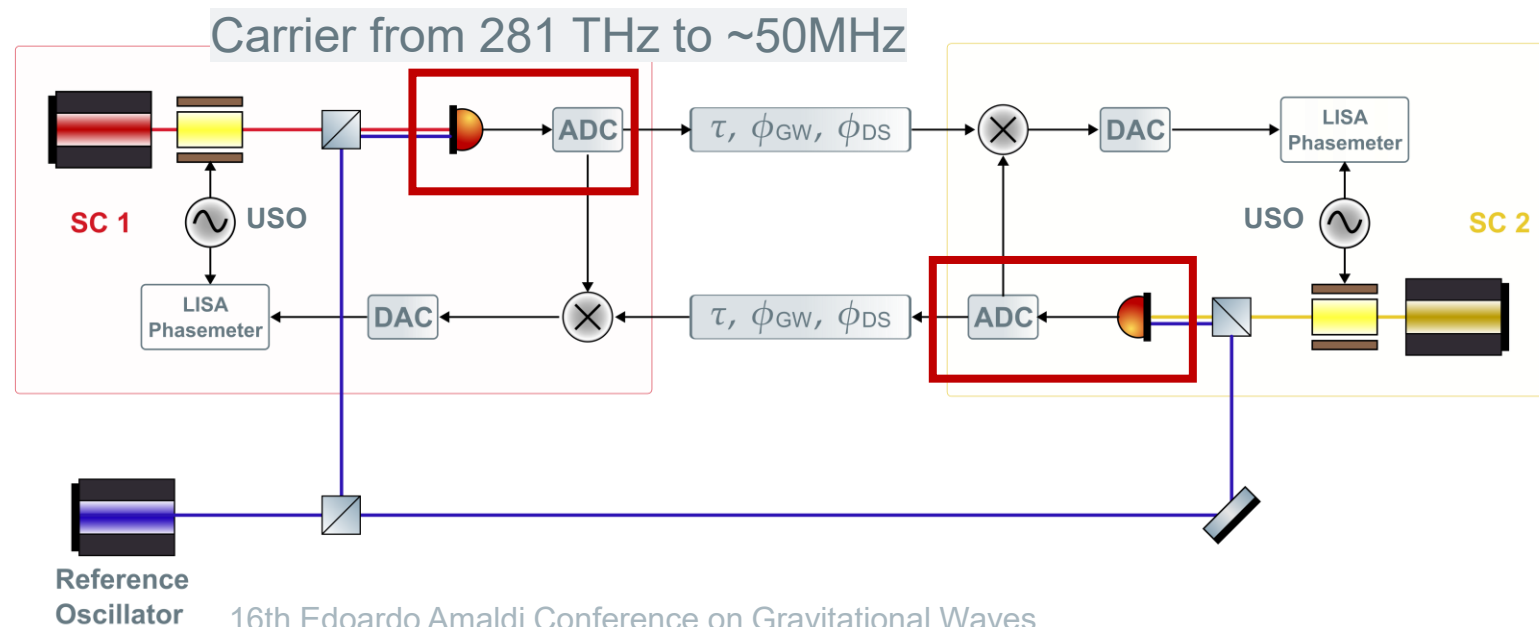
One-arm Phase Measurement



LISA arm



testbed arm



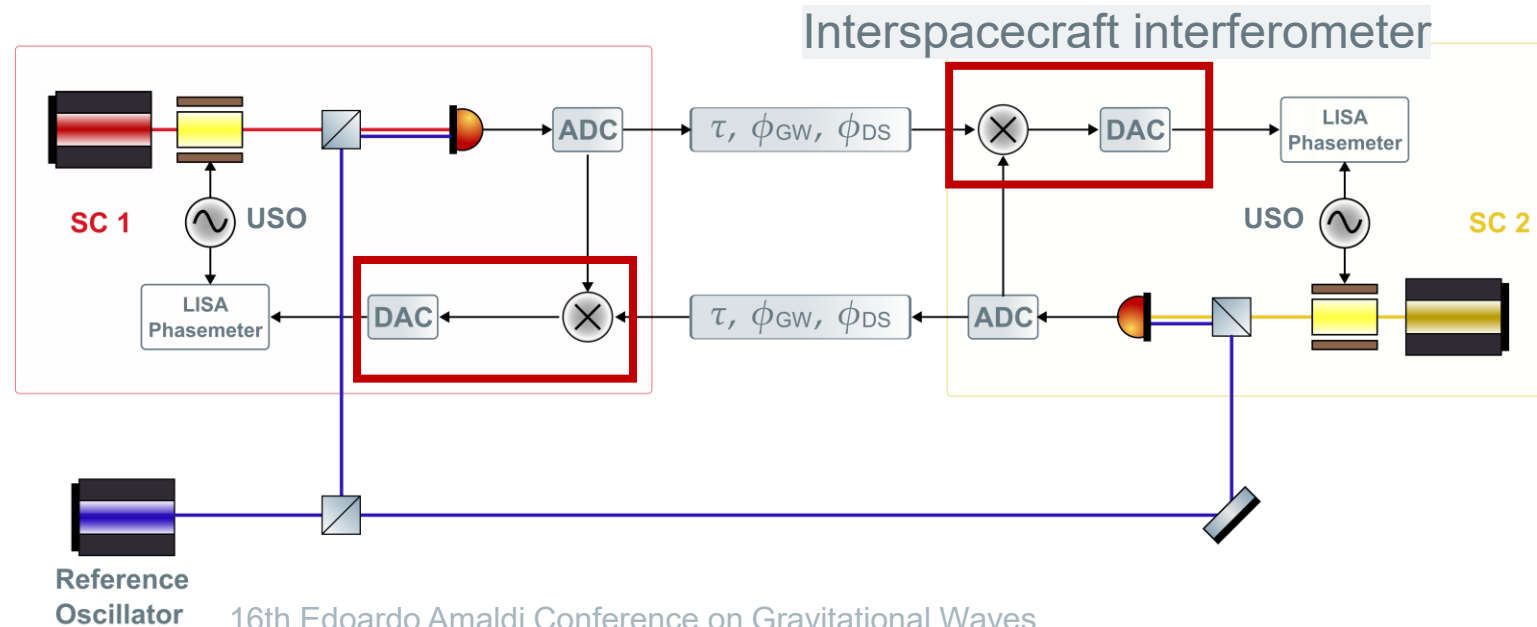
One-arm Phase Measurement



LISA arm



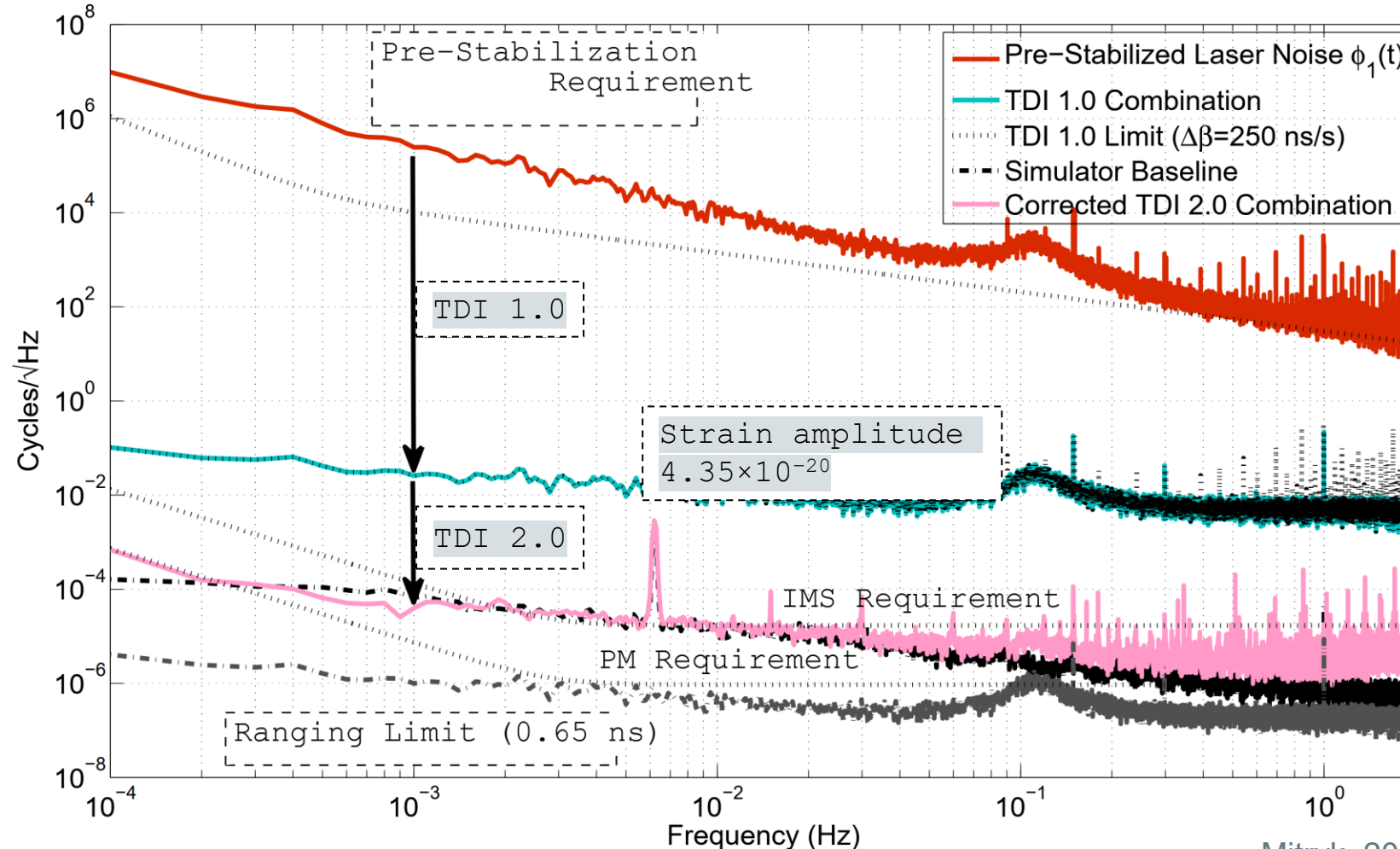
testbed arm



Predecessor Testbed



University of Florida Laser Interferometry Simulator

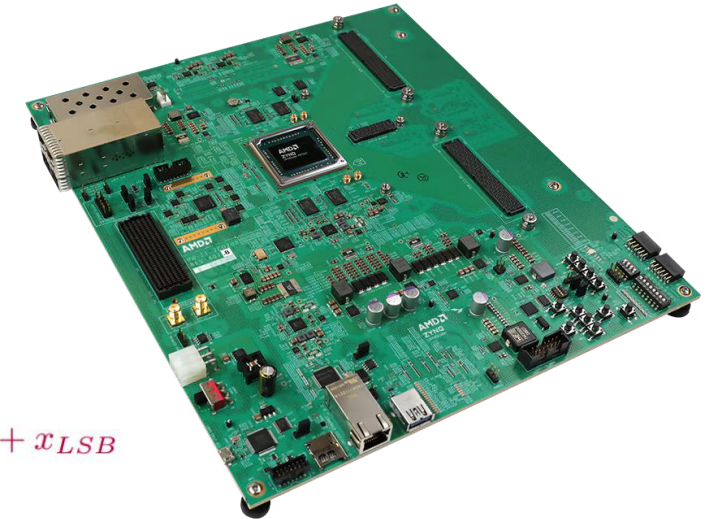
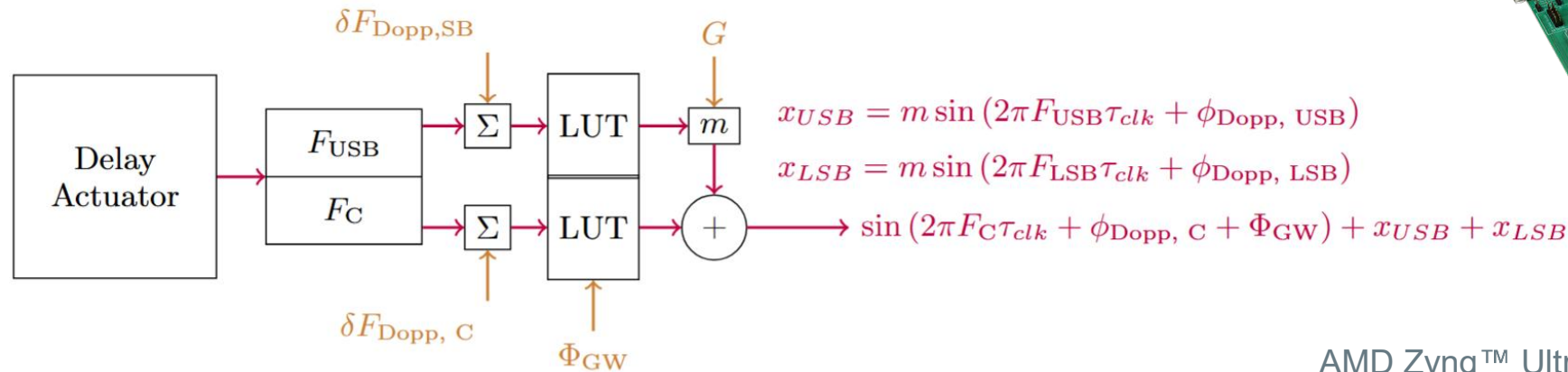


Current Status – Delay Line



A fully electronic implementation of the interferometry system together with a delay and gravitational wave injection has already been developed and is being tested.

- Time varying delays
- Doppler shifts for both the carrier and sidebands
- Gravitational wave injection



AMD Zynq™ UltraScale+™ RFSoc ZCU208 Evaluation Kit

Credit: Reid Ferguson

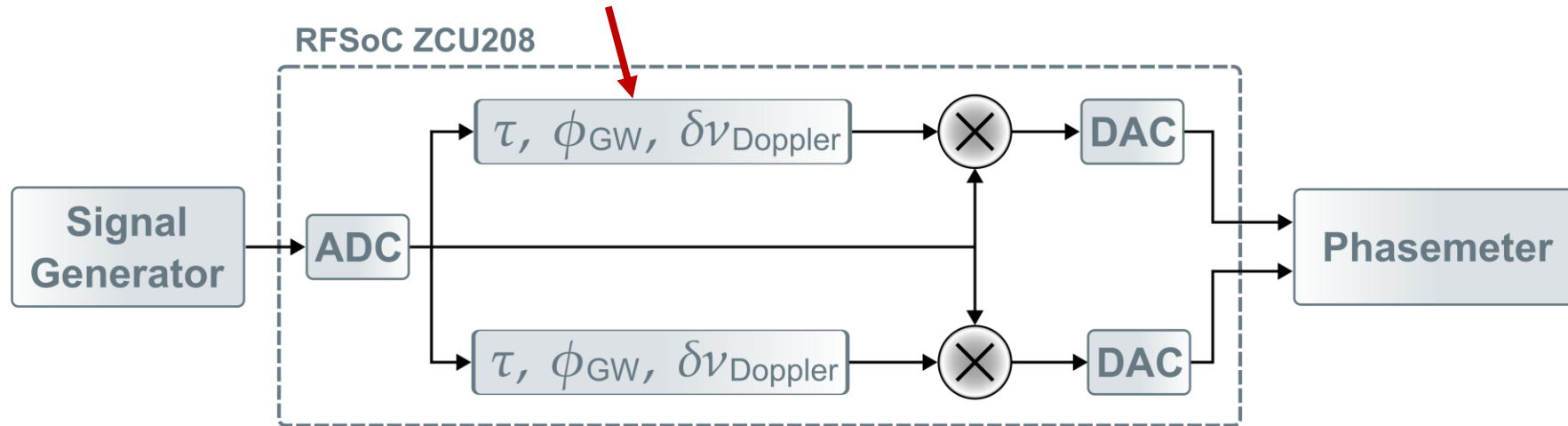
Current Status – Electrical Testbed



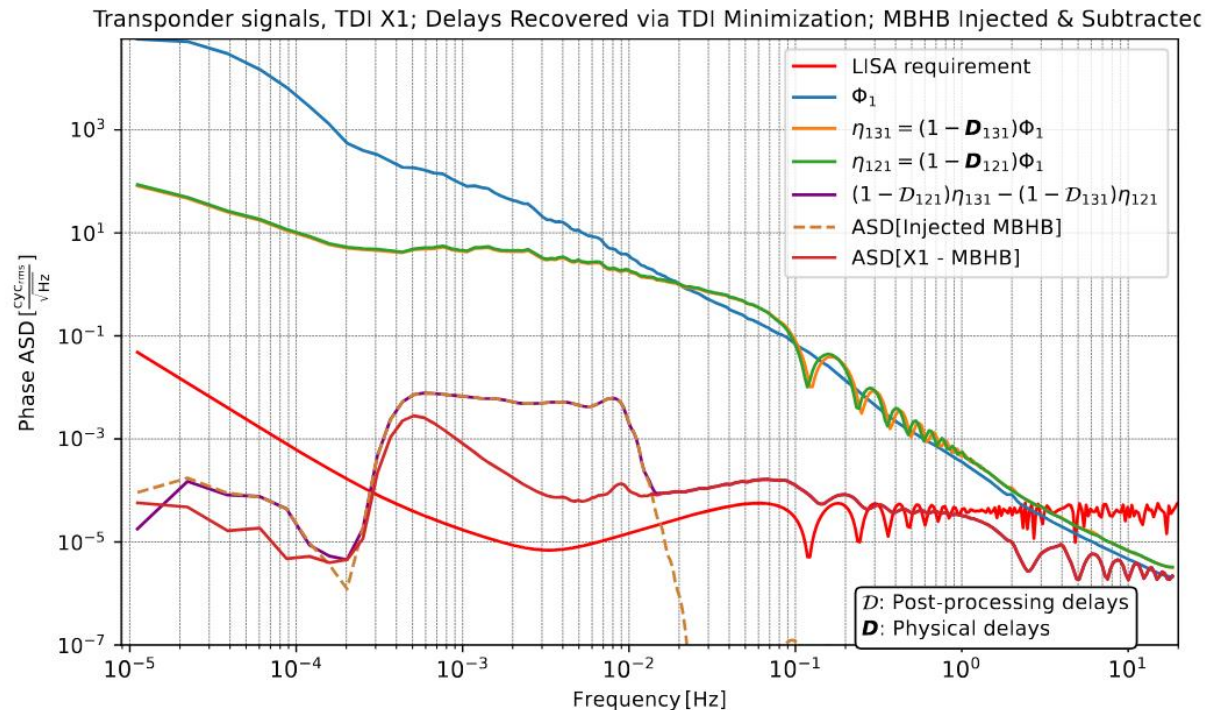
Latest setup for two-arm perfect transponder test

$$X_1 = (1 - \mathcal{D}_{121})\eta_{131} - (1 - \mathcal{D}_{131})\eta_{121}$$

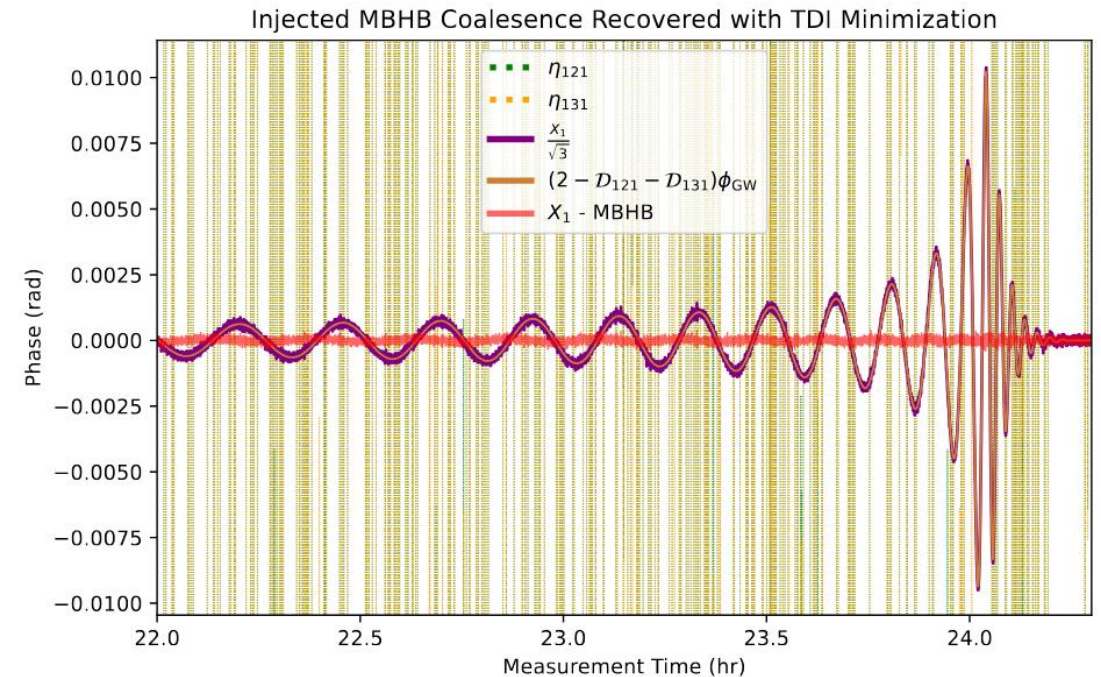
Delays, GWs and Doppler shifts are applied via a on-chip processing system which runs on a separate clock from the rest of the board



Current Status – Electrical Testbed

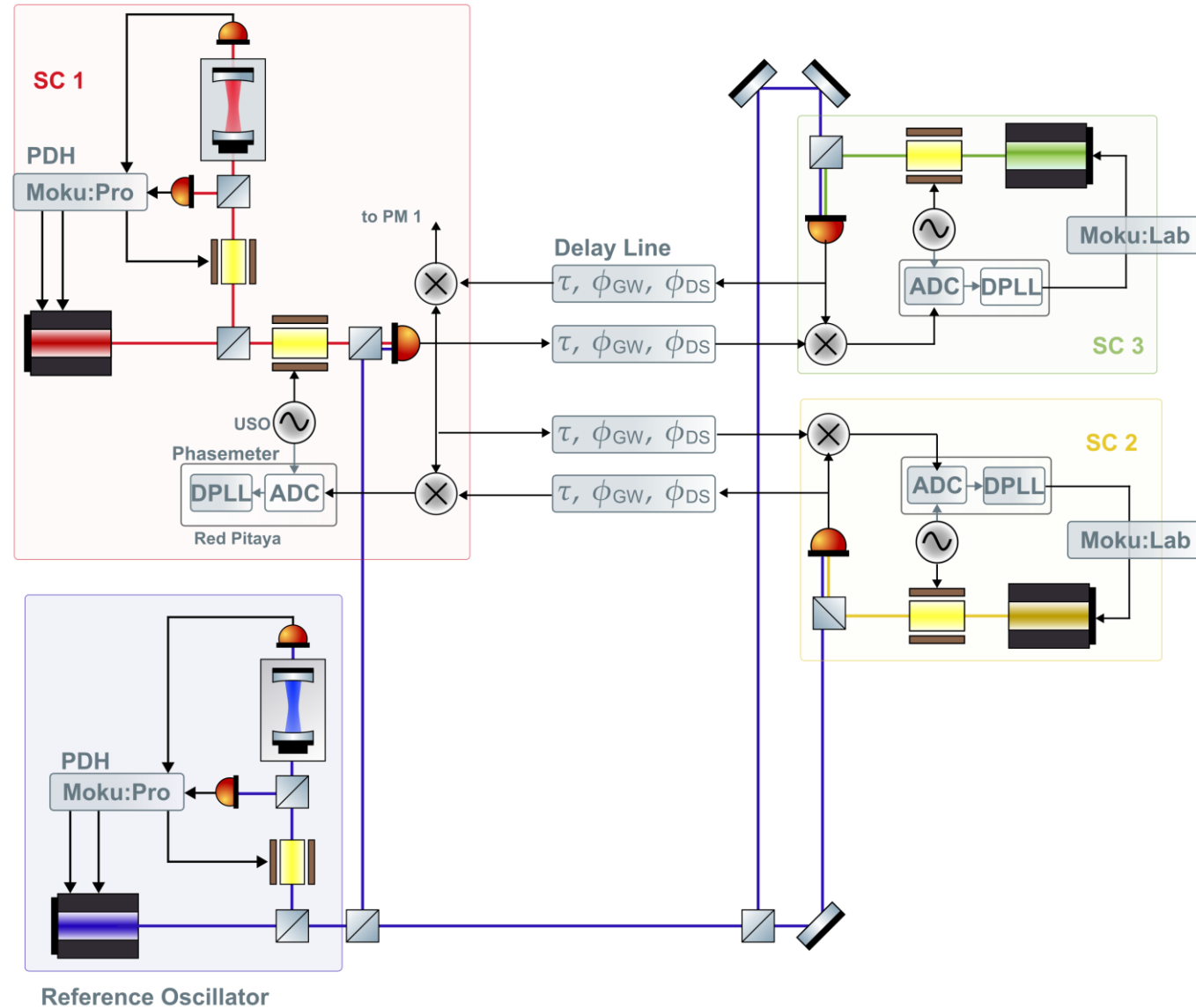


$$X_1 = (1 - \mathcal{D}_{121})\eta_{131} - (1 - \mathcal{D}_{131})\eta_{121}$$



R. Ferguson *et al.*, *Applying TDI to Hardware-Simulated Data*, 2025

Two-arm Testbed with Optical Input



The background is a deep blue space filled with numerous small white stars. A grid of thin, light-colored lines represents the curvature of spacetime, with some lines forming large, dark, irregular shapes that suggest the presence of massive objects or gravitational wells. Several spiral galaxies are visible, glowing with white and light blue light. A prominent, bright red triangle is superimposed on the left side of the image, with its vertices at approximately (50, 300), (530, 550), and (110, 950). The triangle's edges are thick and have a glowing, slightly blurred appearance.

Thank you for listening!